🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.9

Represent Polygons on the Coordinate Plane

Lesson 7-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can draw polygons from their vertex coordinates on the coordinate plane, and use coordinates to find side lengths and solve perimeter and area problems.
Language Objective: I can explain how I found a distance using the words vertices, coordinates, and absolute value.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

vertex | polygon | perimeter | distance between vertices | scale

👆 *Tap any word to see its meaning and a picture.*

1. A corner point of a polygon, named by an ordered pair, is a ________.

2. A closed figure made of line segments is a ________.

3. The total distance around a polygon is its ________.

4. When two vertices share a coordinate, subtract the other and take the ________ ________.

5. What one grid unit stands for in the real situation is the ________.

Write About the Math

Where do the other two vertices go — and how do the known coordinates, the scale, and the area work together to tell you?

¿A dónde van los otros dos vértices — y cómo trabajan juntas las coordenadas conocidas, la escala y el área para decírtelo?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **vertex** *vértice*
- ☐ **polygon** *polígono*
- ☐ **perimeter** *perímetro*
- ☐ **distance between vertices** *distancia entre vértices*
- ☐ **scale** *escala*

Model: *When two vertices share the same y-coordinate, the segment between them is horizontal, so its length comes from the DIFFERENCE in the coordinate that changes — the x-coordinates.* — A strong answer explains that the SHARED coordinate tells you the direction of the side, and the DIFFERING coordinate is what you subtract. A weak answer just says 'subtract the numbers' without identifying which pair.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The park's width is ____ feet. Using the area formula, its length is ____ feet, which is ____ units on the map — so the new vertices sit that many units below the known ones.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **My evidence is ____, so ____.**

  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** vertex
2. **Notes 2:** polygon
3. **Notes 3:** perimeter
4. **Notes 4:** distance between vertices
5. **Notes 5:** scale
6. **Watch for this mistake:** *Subtracting coordinates without the absolute value — from $y = 3$ down to $y = -3.75$ the distance is $3 + 3.75 = 6.75$ units, not $3.75 - 3 = 0.75$. Distances that cross zero are the whole reason the absolute value is there.*
7. **Try It 1:**
8. **Try It 2:**